

```

import math

def entropy(data):
    labels = [row[-1] for row in data]
    total = len(labels)
    counts = {}
    for label in labels:
        counts[label] = counts.get(label, 0) + 1
    return -sum((count / total) * math.log2(count / total) for count in counts.values())

def split_data(data, attribute_index, value):
    return [row[:attribute_index] + row[attribute_index+1:] for row in data if row[attribute_index] == value]

def info_gain(data, attribute_index):
    total_entropy = entropy(data)
    values = set(row[attribute_index] for row in data)
    total = len(data)

    sub_entropy = 0.0
    for val in values:
        subset = [row for row in data if row[attribute_index] == val]
        sub_entropy += (len(subset) / total) * entropy(subset)

    return total_entropy - sub_entropy

def id3(data, features):
    labels = [row[-1] for row in data]
    if len(set(labels)) == 1:

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        return labels[0]
    if len(features) == 0:
        return max(set(labels), key=labels.count)

    best_feat_idx = max(range(len(features)), key=lambda i: info_gain(data, features[i]))
    best_feature = features[best_feat_idx]

    tree = {best_feature: {}}
    remaining_features = [f for i, f in enumerate(features) if i != best_feat_idx]

    feature_values = set(row[best_feat_idx] for row in data)
    for val in feature_values:
        subset = split_data(data, best_feat_idx, val)
        tree[best_feature][val] = id3(subset, remaining_features)

    return tree

# Example Usage
if __name__ == "__main__":
    dataset = [
        ['Sunny', 'Hot', 'High', 'Weak', 'No'],
        ['Sunny', 'Hot', 'High', 'Strong', 'No'],
        ['Overcast', 'Hot', 'High', 'Weak', 'Yes'],
        ['Rain', 'Mild', 'High', 'Weak', 'Yes'],
        ['Rain', 'Cool', 'Normal', 'Weak', 'Yes']
    ]
    features = ['Outlook', 'Temperature', 'Humidity', 'Wind']

    tree = id3(dataset, features)
    print("Generated ID3 Decision Tree:\n", tree)

```

Program 7: Decision Tree Learning using ID3

Generated ID3 Decision Tree:

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{ 'Outlook': { 'Overcast': 'Yes',  
              'Sunny': { 'Humidity': { 'High': 'No', 'Normal': 'Yes'},  
                        'Wind': { 'Weak': 'No', 'Strong': 'No'}},  
              'Rain': 'Yes'}}
```

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